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Safety and efficiency evaluation of an innovative plasma jet array in argon using gas switching technology

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Abstract

Wound healing is an important and still-challenging task in modern medicine. In particular, therapeutic options for treating chronic wounds linked to diseases like diabetes are limited. One promising approach is the application of cold atmospheric plasma (CAP) via medical plasma jets (PJs) or dielectric barrier discharges to specifically stimulate the healing process of non-healing wounds. However, limitations occur regarding the treatment area in the case of PJs.

Thus, an innovative PJ array system (PJAS) was developed to extend the capabilities of established PJs in treating large areas like chronic wounds. Utilizing patented gas switching technology, the PJAS aims to provide effective treatment over larger areas within shorter application times. Determination of performance and safety indicators included measurements of leakage current, temperature, starch–potassium iodide tests, suspension tests, cell viability tests and inhibition zone assays. Values are compared with a medical PJ system, the kINPen® MED.

Findings indicated that the PJAS showed enhanced efficacy compared to present medical PJs. Systematic analyses of leakage current, neutral gas temperature, and emission spectra confirmed the device's safety. A two-step observation was found in the effluent temperature profile of the PJAS with a possible relation to free jet physics and argon-enriched zones. Starch–potassium iodide experiments revealed that the PJAS system oxidizes an area up to twice the size treated by the PJ under the same conditions. Antimicrobial suspension tests indicated an inactivation ranging from 2.1 to 4.9 orders of magnitude, while inhibition zone assays resulted in an up to

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14-fold increase in the effective area for *S. aureus* and to a twofold increase for *C. albicans* in direct comparison to PJ. Additionally, treatment of human fibroblast cells indicated preserved cell vitality for 30 s of plasma exposure. Overall, the PJAS has the potential to become a promising therapeutic option for chronic wound treatment.

Keywords: chronic wounds, medical device, plasma jets, plasma medicine, cold atmospheric pressure plasma, antibacterial efficacy, flow switching technology

1. Introduction

In the last few decades, medical and pharmaceutical research has made outstanding progress in the still-challenging therapy of wounds and also the treatment of diseases like diabetes, which negatively affects or delays the process of wound healing. A multitude of factors have an impact on wound healing. Individually adjusted therapy often involves a combination of different modern wound dressings, hence the application of antibiotics and likewise the treatment of already existing wounds becomes a complex endeavor. The technology of cold atmospheric plasma (CAP) has been found to positively influence the wound-healing process, especially for poorly healing chronic wounds [1, 2].

CAP generates reactive oxygen and nitrogen species (RONS) in the discharge itself, as well as in the surrounding air, in the treated liquid and possibly inside the skin [3]. These RONS play a major role in physiological processes. Thus, studies have shown that an increase in reactive species correlates with an enhanced healing process of wounds [4, 5]. Such RONS occur naturally in the human body in various signal transduction pathways [6]. Furthermore, CAP is also known to reduce the number of bacteria, which is a relevant factor in wound-healing processes [7, 8].

However, the generation and application of plasma can be carried out in different ways, as various geometries and physical options exist. Thus, dielectric barrier discharges and plasma jets (PJs) are commonly used to treat wounds in plasma medicine [2, 9, 10].

The only cold atmospheric pressure PJ approved for medical therapy at the moment is the kINPen® MED, a medical device class IIa [11, 12] since 2013 according to Directive 93/42/EEC and in accordance with the European medical device regulation since 2024. This medical device is intended for the treatment of chronic wounds and pathogen-related skin diseases. The effect of cold atmospheric plasma on wound healing has been intensively studied to follow reaction and physiological pathways involved in the healing process [1].

In general, for the treatment of wounds it is important that the plasma can be applied concisely and adjusted to the respective wound. Moreover, the treatment time should be as short as possible. A PJ offers the flexibility to meet the needed adjustments to the respective wounds. The disadvantages of a sole PJ are the limited treatment area of a few mm^2 [13] and challenges in scalability. For this reason, a PJ array system (PJAS) was developed. The PJAS addresses the need for a time-efficient plasma treatment also for large wounds but retains the flexibility of a single PJ at the same time. Therefore, the PJAS circumvents the aforementioned problems of a single PJ [14, 15].

The device was constructed, based on the experiences made with the established PJ kINPen® MED. Although the small size of the kINPen® MED allows an easy handling and optimal treatment of wounds in defined areas, it has limitations in the treatment of large wounds at an appropriate time, because the treatment time for wound healing recommended by the manufacturer is 30–60 s cm^{-2}. Although first results on this new device are promising for future applications and further studies, especially for the determination of optimal treatment times, distances between the treated surface and nozzles etc. are indispensable. However, the intended use of the device for medical applications requires safety studies.

The PJAS within this study is based on an innovative gas switching technology. This device is supposed to enlarge the treatment area with concomitant faster treatment time compared to a single spot PJ, while posing no risk for the treated patient. Additionally, the PJAS should operate at permanent power instead of a 50% duty cycle like the kINPen® MED. To address these points, a multitude of physical parameters, e.g., neutral gas temperature, patient leakage current (PLC), optical emission spectroscopy (OES), and the formation of reactive species in air were measured. Furthermore, chemical and biological evaluations showcasing the area distribution of reactive species based on starch–potassium iodide test in motion and under static conditions were made, followed by antimicrobial efficacy tests performed in suspension and inhibition zone assays. Additionally, the effect on skin cells (fibroblasts) was determined by cytotoxicity assay. All investigations were based on the German pre-standard DIN SPEC 91315:2014 [16] and the values were compared with the literature or obtained data for the kINPen® MED.

2. Materials & methods

2.1. Plasma device

The upscaling of PJ systems has been a field of investigations in the literature [17–22]. Many of the described solutions are based on a single electronics circuit to drive a helium jet array. However, we determined that the transfer of this circuit to argon discharges generated too much cross talk between the high voltage lines and was thereby not suitable. Another

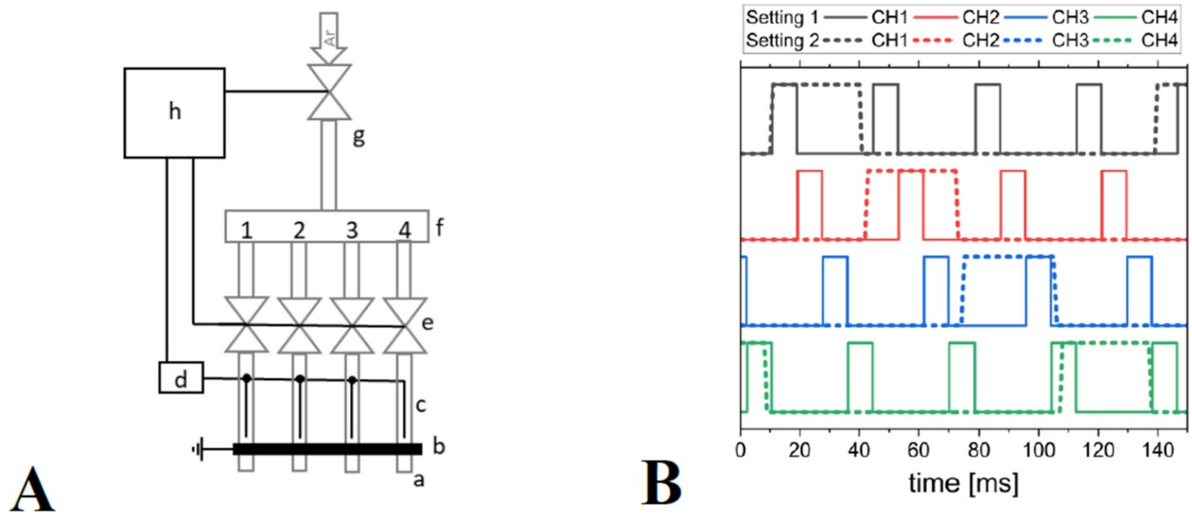

Figure 1. (A) Scheme of the plasma jet array system, with capillary a, grounded electrode b, high voltage needle electrode c, power supply d, gas switching valves e, gas distribution chamber onto channels 1 to 4 f, flow controller g and control unit h, (B) flow scheme for settings 1 and 2 on the all channels showing the electrical signals, high level indicating valve opening.

approach would be to upscale the system by supplying each discharge arrangement with an individual voltage unit. Yet high manufacturing costs and weight counteracted this possibility. The present contribution describes, to the best of our knowledge, for the first time a PJ upscaling via a patented gas switching technology (figure 1 (A) [23]. In theory, the upscaling of this technology is unlimited. Here, an upscaling of the four channels is under consideration for potential future medical applications.

The PJAS implemented the gas switching technology by distributing the gas from a central supply chamber onto individual jet outlets. In this system, each electrode geometry is based on the kINPen® MED setup with a central powered electrode surrounded by a capillary and an outer grounded ring electrode [12]. It has to be noted that the PJAS is a device under development and did not show conformity with any medical regulation yet. An argon (99.999% purity; Air Liquide, Bremen, Germany) flow rate of 2.5 slm up to 5 slm was permanently set at the flow controls and temporarily supplied per channel via quick switching valves, flushing the needle and initiating a discharge in the freshly supplied argon environment (figure 1(B)). While the other needles are still at high voltage, the discharge occurs in the argon environment. The parameters of the valve switching can be adjusted by a controller, but in the present contribution two replicas were operated with one primary setting each:

- Setting 1: a fixed value of 8 ms flushing per channel, resulting in programming delay of 8.2 ms up to 8.3 ms for the physical measurements
- Setting 2:30 ms valve opening and 2 ms delay before consecutive next valve opening.

Given the short flushing time and respecting the inertia of the gas flow (figure 1(B)), the argon started diffusing before a new gas pulse was initiated to replenish argon in the discharge arrangement. The real time duration of discharge ignition at each channel is defined via further conditions like inertia of the working gas succeeding the valve opening according to the settings. The working gas was further subjected to pulling forces from nearby valve flushing and diffusion. Also, delays in the control programming language, mechanical valve opening function and the gas passage through the tubing had to be considered. The flow through the discharge arrangement is presently a topic under investigation [24]. A detailed study of the impact of the valve parameter on the discharge duration and performance will follow in a later stage of development.

The PJAS system was operated with the kINPen® MED power supply (3 kV peak to peak voltage at 0.9 MHz frequency, neoplas med, Greifswald, Germany) without burst. The flow rate was controlled in two stages: within the individual labs via a mass flow controller (physics: F-201 CM with E-8412-A-10, 5 slm range, Bronkhorst, NL; biology: multi gas controller 647C, 20 slm range, MKS instruments, Andover, USA) and within the power unit via a built-in flow control. In the present investigations, two PJAS systems were assembled and employed within this study. Slight changes occur during the successive assembly of each device, with the numbers equaling the appearance within the manuscript:

- Device I: having a homemade integrated circuit for the gas switching technology and the high voltage circuit receiving 66 V DC voltage from the base unit
- Device II: having a homemade discrete circuitry for the gas switching technology and the high voltage circuit receiving 60 V DC voltage from the base unit.

The final combination of devices and settings together with association to respective measurements is listed in table 1. In

Table 1. Devices and gas switching settings used for individual diagnostics within this manuscript with PLC describing the patient leakage current, OES the optical emissions spectroscopy and ROS the visualization of reactive oxygen species distribution via the starch–potassium iodide test.

Device	Gas setting	Flow rate (slm)	Diagnostics applied
PJAS—I	1	2.5	Temperature, PLC, OES
		5.0	Temperature, PLC, OES
	2	2.5	Temperature, PLC
		5.0	Temperature, PLC
PJAS—II	2	2.5	ROS, cell viability, antimicrobial efficacy
		5.0	Temperature, PLC, reactive species
PJ	None	5.0	Temperature, reactive species, ROS, antimicrobial efficacy

the following chapters, the captions will mention the devices as e.g. 'I-2' for device I at setting 2.

In addition, a commercial kINPen® MED device (PJ) was used for comparative measurements, connected to the same gas flow supply with an argon flow rate of 5 slm. For the PJ, it is already known that a far or close distance of the effluent to the surface causes a respective mode shift from free to conductive, determining the influence of the species production on the surface and thereby the treatment efficiency of the two modes [25]. However, practical experiments show that the efficacy is also dependent on device settings and surface properties. Thus, the surface permittivity and conductivity showed an impact on the discharge behavior [26, 27]. In this study, it was not possible to define a single distance between respective modes for all investigations due to the different kinds of measurements, e.g., on copper surfaces for physical device characterization or on agar surfaces for biological tests.

2.2. Device characterization

The device was characterized for the first time with a focus on its efficacy and safety by temperature, PLC, light-emitting species and the output of long-living reactive species. These measurements were designed to allow an evaluation for a potential medical application and were thus linked to the DIN SPEC 91315:2014 [16, 28], yet this should not exclude the technology from further fields of applications.

2.2.1. Temperature measurements. In order to acquire the temperature as a function of distance, a fiber-optical temperature (FOT) probe (0.2 K precision, 0.55 mm out diameter, model TS5, Weidmann, Germany) was placed in front of each channel 1 to 4 in succession, oriented to face away from the gas flow direction. The distance was controlled either with a motorized stage (MTS50/M-Z8 in the *XYZ* configuration with TSC 001 motor control, Thorlabs, USA) controlled with Kinesis (Version 1.14.37, Thorlabs, USA) for PJAS I-1 or manually with a linear stage for PJAS I-2 and II-2. The FOT probe was placed in front of each nozzle to measure the core of the outlet flow as well as sideways for radial profile measurements.

Despite previous measurement routines, the present detection system was operated at constant movement and constant data acquisition of the FOT probe. Thereby, a more detailed profile of the temperature as a function of distance could be resolved [29]. For the temperature profile over distance, the motorized stage moves at a constant velocity of 0.15 mm s^{-1} with the FOT probe moving either constantly away or toward the capillary orifice. The profile for each channel of the PJAS as well as the PJ was acquired. For the overall temperature map of the PJAS, the FOT probe also moved continuously sideways over the full array of channels with a velocity of 0.5 mm s^{-1}. The distance to the outlet was stepwise increased at 1 mm resolution.

2.2.2. Patient leakage current. In order to determine the PLC, a similar operation of the motorized stages was used. A copper strip on a plastic surface serves as a counter electrode. The counter electrode was connected via an RC-circuit (according to DIN EN 60 601-1-6 via UNIMET® 800ST, Bender GmbH & Co. KG, Gruenberg, Germany) to ground [30, 31]. The motorized stage moves the grounded electrode toward the capillary orifice at a velocity of 0.1 mm s^{-1} from a distance of 10 mm toward 1 mm. The measurements were performed for all four channels simultaneously and for individual channels as well. For the latter setting, the grounded electrode is shielded with a plastic part except for an access area of 1×1 cm^{-2} in front of the investigated channel.

Data acquisition for temperature and PLC measurements was performed with the homemade Python code PlaDinSpec (Python 3.8).

2.2.3. Optical emission spectroscopy. OES was employed to identify the emitting species as a function of distance by the PJAS. All OES measurements were performed with the PJ impinging on a grounded metallic mesh over a quartz plate

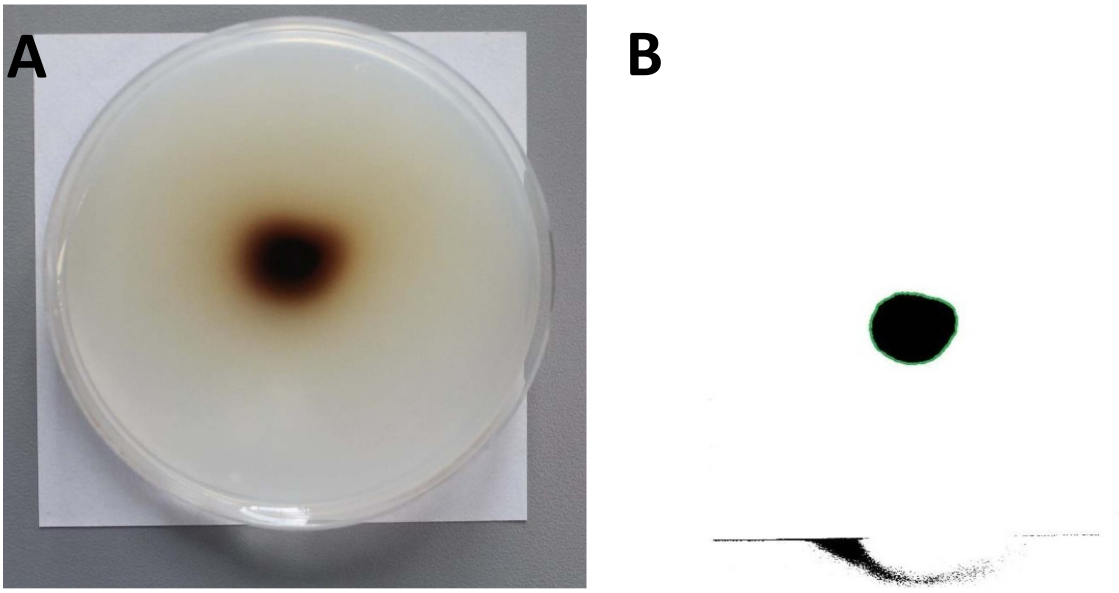

Figure 2. Example of an image before transformation (A) and the same image transformed for the measurement of the effective treatment area (the effective area represented by a green boundary; (B).

similar to a setup described previously [32]. A spectrometer (Avantes, AvaSpec 3648, 200–1100 nm range, ∼2 nm resolution, 300 line mm^{-1} grating, blazed at 300 nm, deep-UV-detector coated CCD linear array) with an optical fiber connected to a cosine corrector was employed for OES measurements. The optical fiber was placed in front of each PJAS channel in succession at 80 ms integration time and 100 averages. The OES system was calibrated for absolute light intensity measurements. The distance between the plasma outlet and target was varied from 3 mm to 25 mm.

2.2.4. Reactive species measurements. Long-lasting reactive species in the gas phase were measured using a commercial gas detector system (model APOA-360 for O_3 and APNA-370 for NO_2, Horiba, Japan). The PJAS was placed in front of the detection tube (end on, 0°) at a distance of up to 100 cm. In addition, measurements were performed with the PJAS at an 90° impact angle on a dielectric plate at 10 mm distance and the detection tube placed at an angle of 45° or 90° toward the PJAS at a distance of up to 100 cm away (see further [28]).

2.3. Visualization of reactive oxygen species (ROS)

2.3.1. Starch–potassium iodide test. For the determination of the plasma effective zone, the formation of ROS was analyzed using a mixture of starch, potassium and iodide, as described previously for the kINPen® MED [33]. In the presence of ROS, iodide ions were oxidized to iodine, which was described to be an indicator for agents like ozone, hydrogen peroxide or nitrous acid [34]. Iodine further reacted with the iodide species to polyiodide anions, which were then intercalated with amylose helices from the starch. As a result of these complexes, the color changed from white/cream to purple/brown. Plates were prepared according to the instruction described by Nishime *et al* [35]. The used circular petri dishes possessed a diameter of 11 cm and each plate was filled with 50 ml of the prepared starch–potassium iodide agar mixture.

The analysis of the plasma-treated plates was performed by measuring the dark colored area using the software ImageJ (US National Institutes of Health, Bethesda, MD, USA). Therefore, the images were first scaled using the known diameter of the plate to define the accurate pixel to mm ratio. Afterwards, all images were transformed to 8-bit grayscale and the thresholds of these grayscale images were adjusted using the 'MaxEntropy' method of the 'Auto Threshold' function. The effective treatment area was measured after selecting the black-ish area in the plate center with the tracing (wand) tool and using the 'Measure' function in the 'Analyze' menu.

An example of the visualization after the ImageJ transformations is shown in figure 2. The evaluated area is limited to the image center region (framed in a green circle for display).

2.3.2. Experimental setup. To imitate the movement of the devices, a *XYZ* translation stage was positioned and moved along individual predefined pathways. A traversing path on the starch–potassium iodide plates was used for the PJAS and comprised a movement path with four lines (figure 3). For the PJ, the same settings regarding path distance and operation length were applied as for the PJAS. The only difference was the programming of the *XYZ* stage, to run the four lines separately. The treatment distance for both devices was set to 10 mm.

To investigate the impact of setup variations (number of movements and movement speed) on the resulting effective

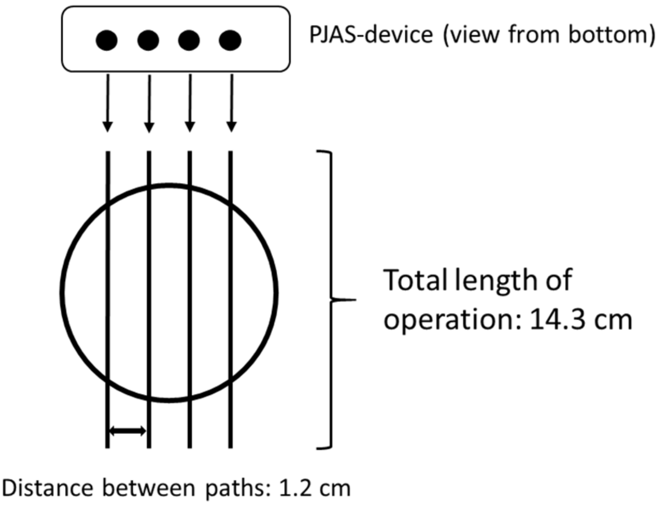

Figure 3. Scheme of the pathway pattern of the PJAS.

area, the following experiments were performed for the PJAS ($n = 3$ each):

(1) PJAS moved once over the length of operation.
(2) PJAS moved four times over the length of operation.

The PJ was moved four separate times, to obtain four individual lines (analogous to the pattern in figure 3). Regarding the gas flow settings, both devices were operated with argon, while the flow rate of the PJAS was adjusted to 2.5 slm and 5.0 slm for the PJ, as recommended by the manufacturer. In order to evaluate the influence of the speed of operation on the respective effective area, three different speed rates (0.125 cm s^{-1}, 0.250 cm s^{-1} and 0.500 cm s^{-1}) were tested. Controls were carried out for the lowest speed rate (0.125 cm s^{-1}) with both setups for the kINPen Multijet and the kINPen® MED.

2.4. Cell viability test

Cell viability tests were performed following the DIN SPEC 91315:2014 [16]. Therefore, 96-well plates were used as the distance between the nozzles of the PJAS are consistent with the distance between each well of a 96-well plate. Hence, each well is treated by one nozzle of the PJAS. The settings on volumes were attuned and a cell number of 9.1×10^3 cells were seeded per well of a 96-well plate. Cell viability was assessed using a resazurin-based alamar assay (alamarBlue, Thermo Fisher Scientific, Dreieich, Germany).

For the treatment, an adherent human skin fibroblasts cell line GM00637 (immortalized by the Simian Virus 40 transformations) was used (Coriell Institute, Camden, New Jersey, USA). Cells were cultured at 37 °C (5% CO_2, 90% humidity) in Dulbecco's modified Eagle's medium (DMEM) high glucose with L-Glutamine (Coelbe, Germany), supplemented with 1% penicillin/streptomycin (Coelbe, Germany) and 8% fetal bovine serum (FBS; Biochrome AG, Berlin, Germany). The fibroblasts were divided twice a week in 175 cm^2 cell culture flasks (Greiner Bio One GmbH, Frickenhausen, Germany).

Before plasma treatment, the cells were seeded into the wells of a 96-well plate and incubated for 6 h. After incubation, the medium was removed and the cells were washed twice with 50 µl of phosphate-buffered saline (PBS, pH 7.4). Cells were treated with the PJAS for 10 s, 30 s, 60 s, 120 s, and 180 s ($n = 3$). The distance was set to 10 mm between the nozzle and the liquid surface. Directly after each treatment, 150 µl of fresh DMEM with 13% FBS was added to the cells.

Next to the treatment with the PJAS, a positive control was carried out by using a H_2O_2 solution with a concentration of 0.00025% (7.5 µM) [16], to serve as a model for the ROS, which are mainly responsible for biological plasma effects [36, 37]. The negative control was performed by the addition of DMEM only.

After completing the experiments, the cells were incubated with resazurin at a total concentration of 100 µmol l^{-1}, for 4 h at 37 °C and 5% CO_2. Afterward the fluorescence was measured using a multiplate reader (Tecan F200, Tecan Group Ltd, Männedorf, Switzerland) at $\lambda_{ex} = 560$ nm and $\lambda_{em} = 590$ nm.

2.5. Antimicrobial efficacy tests

2.5.1. Microorganisms and culture conditions. According to the DIN SPEC 91315:2014, the following five microorganisms (DSMZ-German Collection of Microorganisms and Cell Cultures, Braunschweig, Germany; ATCC-American Type Culture Collection, Manassas, USA; NCTC-National Collection of Type Cultures, Chelsea, UK) served for the determination of the antimicrobial efficacy of the PJAS in comparison to the PJ [16] and were purchased from the German Collection of Microorganisms and Cell Cultures, Braunschweig, Germany:

- *Staphylococcus aureus* DSM 799/ ATCC 6538
- *Staphylococcus epidermidis* DSM 20 044/ ATCC 14 990
- *Escherichia coli* K-12 DSM 11 250/ NCTC 10 538
- *Pseudomonas aeruginosa* DSM 50 071/ ATCC 10 145
- *Candida albicans* DSM 1386/ ATCC 10 321

The bacteria *S. aureus, S. epidermidis, E. coli* and *P. aeruginosa* were cultured in 20 ml of soybean casein digest broth (Carl Roth GmbH & Co. KG, Karlsruhe, Germany) at 37 °C for 24 h, whereas the yeast *C. albicans* was cultured at 30 °C for 48 h. After incubation, 10 ml of the culture was centrifuged for 300 s at 4500 rpm (Heraeus Multifuge 1 S, Thermo Fisher Scientific, Dreieich, Germany). The pellet was then resuspended in 10 ml of 0.85% physiological sodium chloride solution. The resulting suspension was diluted 1:1000 to obtain a total viable cell count of approximately 10^5 colony-forming units (cfu) ml^{-1}. This solution was used for the suspension test and the inhibition zone assay, which were performed in accordance with the DIN SPEC 91315:2014 [16].

2.5.2. Suspension test. Tests were carried out in petri dishes with a diameter of 55 mm, since the PJAS is wider than the PJ due to the four nozzles. The argon flow of the PJAS was adjusted to 2.5 slm. A volume of 5 ml was treated. The distance between the nozzles and the surface of the suspension was set to 5 mm. Treatments of 60 s, 120 s, 180 s, 240 s and 300 s ($n = 3$) were performed.

After the respective treatment time, samples were plated on soybean casein digest agar (Carl Roth GmbH & Co. KG, Karlsruhe, Germany) plates with a distribution of 50 µl by a spiral plater (Eddy Jet 2, IUL, Barcelona, Spain) to evaluate the total viable count (cfu ml^{-1}). The plates were incubated for 24 h at 37 °C, except for the microorganism *C. albicans*, which was incubated for 48 h at 30 °C. Colony-forming units were afterward counted by an automated colony counter (Flash & Go, IUL, Barcelona, Spain). For the controls, argon gas was applied on the suspension for a maximum treatment time of 300 s.

The reduction of the number of colony-forming units (cfu) per ml is expressed as $\log_{10}(N_R)$. Therefore, the cfu after treatment ($\log_{10}(N_S)$) is subtracted from the cfu of the control (only gas flow ($\log_{10}(N_0)$)).

2.5.3. Inhibition zone assay. Petri dishes (diameter: 90 mm) containing 20 ml soybean casein digest agar were used for the inoculation with the respective microorganisms. Thus, 100 µl of the diluted culture (2.5.1) was spread manually on the plates. Afterward, the plates were spot-like treated by the PJAS at a distance of 10 mm between the nozzle and the agar surface with an argon flow rate of 2.5 slm. Treatments were carried out for 60 s, 120 s, 180 s, 240 s, and 300 s ($n = 5$). For the control, the longest treatment time of 300 s was selected and an application with argon flow without discharge performed.

Afterward the plates were photographed (Canon EOS, 7D, Tokyo, Japan) and the pictures were analyzed by ImageJ. A systematic analysis of the inhibition zones of the various microorganisms was not possible in the same structured way as the area determination on the starch–potassium iodide plates (2.3.1). The reasons for this are the uneven growth of the bacteria, the low-contrast differences between agar and colonies and the reflections and shadows on the plates during photography. Therefore, only a slight image pre-processing was carried out by transforming the scaled color images into 8-bit grayscale images with a subsequent variance filter (radius ranges from 10 pixels to 100 pixels depending on the microorganism and plate growth). The edge of the inhibition zone was then traced on the processed images using the 'Freehand selections' tool and afterward measured using 'Measure' in the 'Analyze' menu.

3. Results and discussion

3.1. Physical characterization

3.1.1. Temperature measurements. In order to characterize the PJAS based on thermal impact on a treated surface comparable to living tissue, the temperature of the neutral gas was determined as a function of distance for both settings. The temperature was measured along the channel outlet of the device without a surface disturbing the flow or the device. Since the device is in a prototype stage, the efficacy per channel can vary. Therefore, a channel specific characterization was performed along with the basic safety and efficacy measurements. The presently used method with the moving probe differs from previously acquired temperature measurements for the PJ (kINPen® MED) [25, 28, 38], hence a new set of data with the identical detection system was acquired.

The neutral gas temperature of the PJAS was determined for two gas flow rates supplied to the pulsed jet technology to have comparable values compared to the biological tests requiring low flow rates and the classical PJ values. Figure 4 shows group temperature slopes per gas flow rate, whereas especially channel 4 (CH4) shows the highest neutral gas temperature for all investigated channels. For the 2.5 slm argon flow rate, the temperature at the outlet ranges from 65 °C up to 75 °C with a rapid drop below 40 °C at 9.5 mm away from the device. For the 5 slm argon flow rate, the temperature ranges from 56 °C up to 62 °C at the outlet, with a drop below 40 °C already at 7.5 mm. CH4 shows temperatures around 2 °C to 10 °C above the other channels, indicating an increased power dissipation into the plasma in this channel [38]. Based on the device geometry, this might be a result of a more centric needle positioning compared to other channels. In the case of the PJAS II-2 with a flow rate of 5 slm, the neutral gas temperature does not reach above 37 °C. A comparison of the four channels shows a weak yet comparable spread with CH4 having the highest temperature. PJAS I has an overall higher temperature compared to PJAS II.

For PJAS I-1, each channel was measured individually at gas flow rates of 2.5 slm and 5 slm. In addition to the overall drop in temperature with distance, multiple temperature steps are observed for both gas flows in all channels. For 2.5 slm there is a step range from 0.7 mm up to 2.9 mm, while for 5 slm a step range from 1.2 mm up to 3.1 mm was measured. Previous measurements in PJ devices over distance did not show such step-wise behavior [28, 38], while a slope change was observed at higher spatial resolution for different power settings [39]. In our present investigation, the temperature within the effluent of the PJ shows two steps as well (figure 4(C)). The overall temperature ranges from 32 °C down to room temperature over the investigated range. The first step occurs at 5 mm and resembles previous observations [39]. This first step position correlates with the previously detected end of the potential core of the free jet, when the emitted gas starts to mix with ambient air due to turbulent gas features [40]. As the percentage of admixed ambient species increases with distance, metastable atoms are quenched by collisions with nitrogen molecules and all remaining energy is transitioned into rotational and vibrational levels acting as a heat source. With the second temperature step at 15 mm supposedly all metastable atoms are quenched as well as the heat diffusion not being of relevance anymore. With these two interpretations of the observed temperature steps in the PJ case, the PJAS case is more complex to discuss. While the PJ operates as a constantly supplied free jet, this is not valid for the PJAS with the pulsed flow setting. The free jet will not be fully developed within the short valve opening time of 8 ms [29]. The consecutive

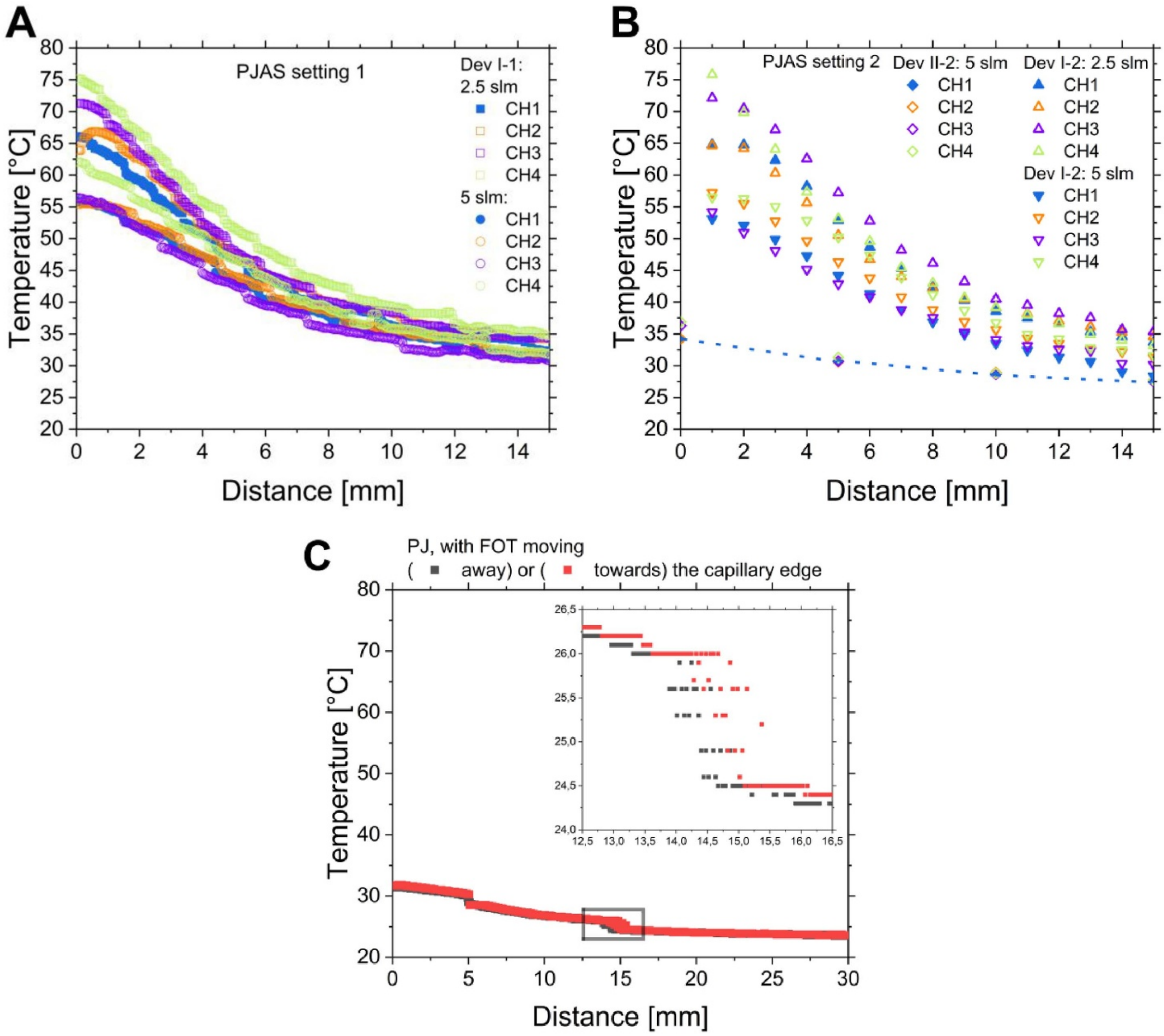

Figure 4. Neutral gas temperature in the effluent by FOT probe measurements for (A) PJAS I-1 and (B) PJAS I-2 as well as II-2 as a function of distance per outlet channel and (C) for PJ as a function of distance per FOT movement direction.

openings of each channel emit multiple argon density enriched zones in a row. Each zone starts to diffuse before a reopening occurs. Considering that the temperature steps in figure 4 have a different characteristic length per flow rate but not per channel, one interpretation would be that each plateau shows a previously emitted argon zone moving away from the remaining impulse, which is a function of the flow velocity and hence the flow rate, as well as being pushed away by consecutive valve openings.

The 2D temperature distribution for 2.5 slm and 5 slm shows a temperature drop with a distance up to 50 mm away from the PJAS orifice (figure 5). For both gas flow rates, even at a z-distance of 50 mm in central position (at $x = 25$ mm), a definite temperature above the ambient temperature of up to 30 °C for 2.5 slm and up to 28 °C for 5 slm was detected. Furthermore, the individual heat profiles generated by the plasma ignition per channel show a tilt from the outer edges of the arrangement to the middle, with the temperature maximum bending up to 2.5 mm toward the center (at $z = 30$ mm). By sequentially flushing argon through different channels, ambient air is dragged in behind the argon enriched zone from all directions. The resulting dragging force overlaps in the central direction. Previous studies of helium have already shown the working gas mole fraction to merge toward the center at constant gas flow [19].

3.1.2. Patient leakage current. The PLC for the PJAS was determined as a function of distance for all channels in both settings as well as individually for PJAS I-1 (figure 6). For a 2.5 slm and 5 slm argon flow rate, PLC was measured with the counter electrode placed 10 mm away and moving toward the PJAS I. For 2.5 slm, the PLC value rose slowly upon the electrode moving toward the discharge effluent to about 0.04 mA until at a distance of 7.7 mm the plasma started connecting to the target electrode and changed the discharge mode—the previously titled conductive mode [25, 41]. With a further reduction of the distance, the PLC quickly rises up to 0.3 mA before dropping again at a distance of 6 mm to values below the PLC threshold of 0.1 mA according to IEC 60 601–1 [42]. At closer distances, the value keeps dropping. At a flow rate of 5 slm, the

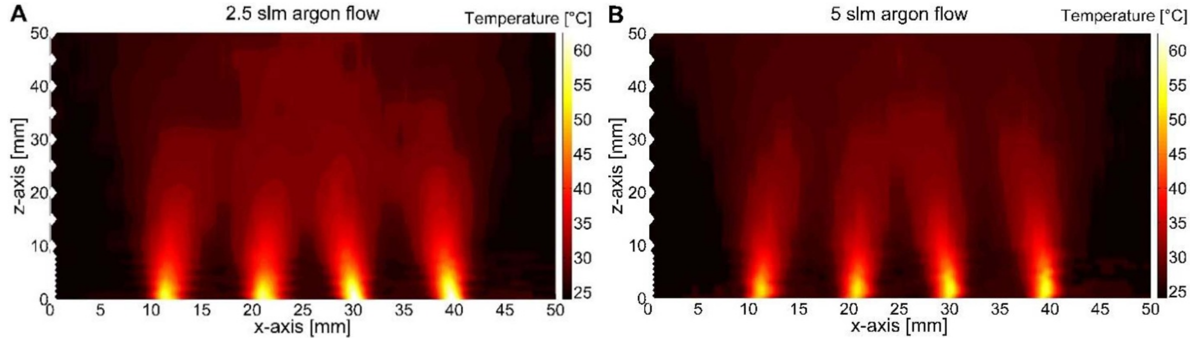

Figure 5. Neutral gas temperature in the effluent of the PJAS I-1 by FOT probe measurements. The probe was moved constantly in x-direction at the indicated z-positions (white dots). Measurements are shown for gas flow rates of 2.5 slm (A) and 5 slm (B).

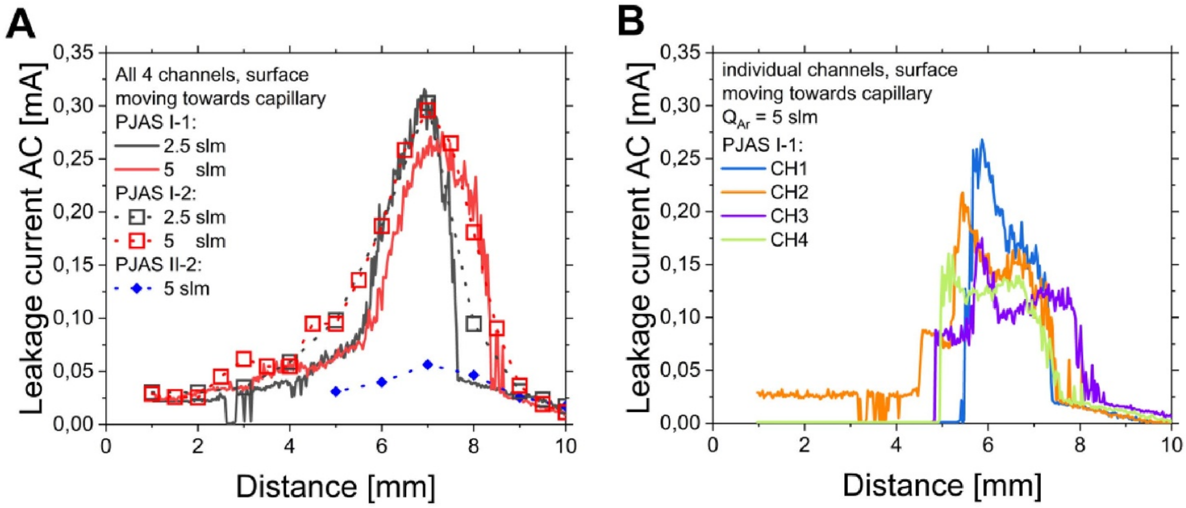

Figure 6. Determination of PLC for the PJAS I-1, I-2 and II-2 at different distances for all channels simultaneously and two flow values (A) and channels independently at 5 slm at PJAS I-1 (B).

conductive mode is already initiated at 8.7 mm and reaches a PLC peak value of 0.27 mA. Moving closer, around 6 mm safe PLC operation values below the threshold of 0.1 mA were reached again. Considering the present device settings, such as valve parameters, applied voltage amplitude and selected gas flow, these measurements indicate a safe operation setting to be at a treatment distance of 8 mm or 9 mm respectively. Also, in the case of a distance sensor implemented with a feedback loop to the high voltage unit [33, 43], a treatment at a close distance smaller than 6 mm could be established in a safe manner.

The comparison between the different valve settings with PJAS I showed no difference for the PLC. For biological investigations, the PJAS II-2 with a reduced input DC voltage as well as a different gas switching electric circuit showed a reduced PLC between 0.06 mA at 7 mm and 0.02 mA at 10 mm distance for 5 slm (see figure 6(A)).

A comparison of the detected values with previous values for the PJ shows a difference in overall shape for 5 slm. In the previous investigation, the PLC dropped continuously with increasing distance [28]. In a further investigation on a research-based kINPen® device without burst (no 50% duty cycle) at a lower gas flow of 1 slm, a local maximum occurred around 7 mm [25]. Another argon-based PJ device also showed the existence of such a local maximum around 7 mm for a gas flow rate of 2.3 slm [44]. Since the PJAS system at 5 slm still has a pulsed gas supply, the argon atmosphere in the effluent region is not comparable to a single jet at permanent 5 slm gas flow. The enhanced admixture of ambient air species into the discharge region could induce further quenching, generate further ionic species and potentially enhance the conductivity in this area [45, 46].

Considering the PLC of the individual channels (figure 6(B)) shows an overall similar shape to the joint PLC curve. The peak value per channel differs from 0.16 mA on channel 4 up to 0.25 on channel 1, which is counterintuitive to the temperature data (figure 4). Channel 3 also shows a switch into the conductive mode over a longer range. For channels 1, 3 and 4 at close distances the plasma extinguishes, while channel 2 prevails until the minimal investigated distance. This range of operation in the case of a grounded counter electrode requires further consideration and optimization by adjusting the device settings (voltage and valve parameters) as well as a calibration of the high-voltage electrode placement within each individual channel.

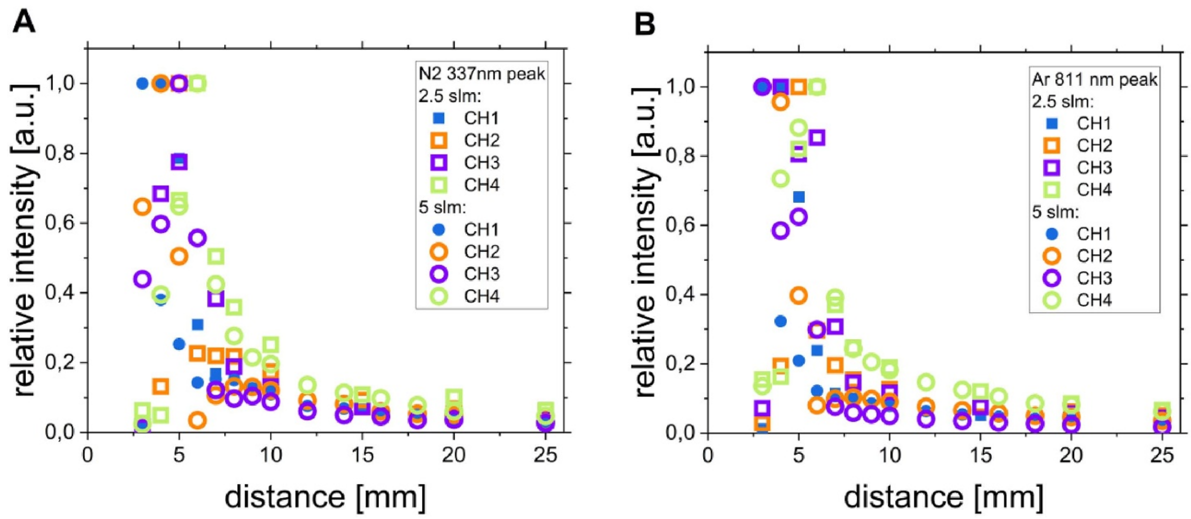

Figure 7. Dependence of relative emission intensity of N_2 SPS 337 nm-peak (A) and representative atomic argon line at 811 nm on distance (B), PJAS I-1 output channel and gas flow rate.

3.1.3. Optical emission spectroscopy. To compare the plasma emitted by each channel, the intensity of the spectral lines was investigated as a function of distance. The overall emission spectra resemble that of the PJ and are hence not shown here [28]. Here, two spectral lines characteristic for PJAS I-1 in ambient air are shown: the 337 nm peak from the molecular nitrogen emission band (second positive system) and the 811 nm atomic argon line (figure 7). The tendency to distance both lines shows similar features and also resembles further evaluated lines, not included here, such as atomic oxygen and the hydroxyl radical. The highest discharge emission occurs around the position where the conductive mode is initiated. It is also observed that the dependence of peak value over distance despite an average of over 100 measurements does not reveal a clear trend over distance for either investigated line. The convolution of valve switching, discharge ignition, prevalence of conductive mode, OES exposure time and OES average does not lead to a homogenous trend.

3.1.4. Reactive species measurements. The production of ozone by the investigated PJAS II-2 was investigated (figure 8) with respect to the DIN SPEC 91315:2014 [16]. For the PJ, the end on 0° production of ozone as a function of distance was measured. In the case of the PJAS, the discharge channels were spread over a wide range of about 30 mm, while the suction tube of the detection device has an opening of 4 mm. As a result, the 0° end-on measurements started at a distance of 20 cm away from the discharge. The detected ozone concentration coincides between both devices. Further measurements with the PJAS in front of the surface revealed values close to the PJ. As a result of this comparative measurement, the ozone production of the PJAS does not pose a concern for a future application, also when linked to previously discussed guidelines [28, 47]. Instead of discussing absolute values acquired with a localized measurement, new routines are under research to qualify the hazardous risk of species production by determining the production rates, thereby allowing an application specific risk assessment by considering room dimensions to conclude the exposure levels.

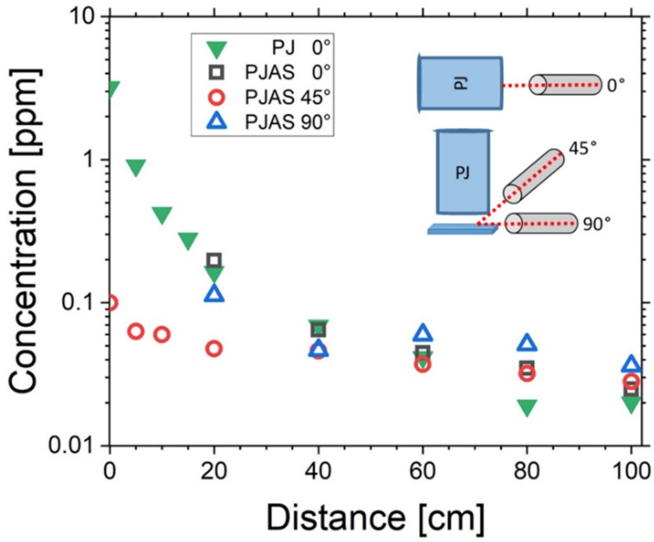

Figure 8. Determination of ozone density in ambient air as a function of distance to the device outlet and the angle of potential exposure for a PJ and the PJAS II-2 each operated at 5 slm argon. In case of sideways detection at 45° and 90°, the PJAS II-2 was placed 10 mm away from the surface.

3.2. Effective area

Experiments with agar plates containing starch–potassium iodide were used for the visualization of the effective surface-impact area of plasma-generated ROS for treatment times of 60 s, 120s, 180s, 240s and 300s as a model for the effective area (figure 9). The effective area was analyzed using ImageJ.

The area increased with rising treatment time. For a treatment time of 60 s the nozzle position of the PJAS was visible in the round spots lined up close to each other. With

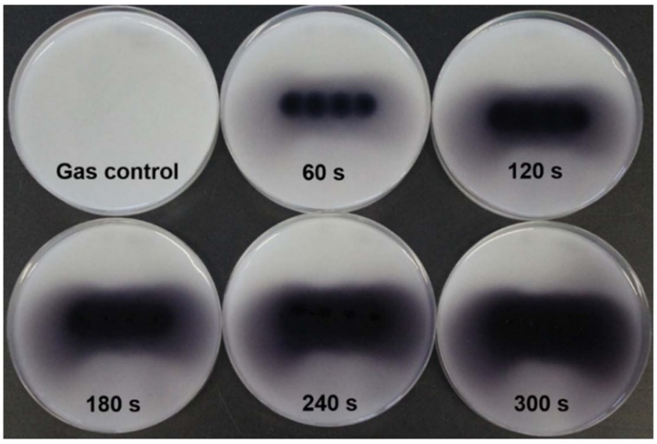

Figure 9. Image of the effective area after 0 s (gas control) and treatment times of 60 s, 120 s, 180 s, 240 s and 300 s for the PJAS II-2.

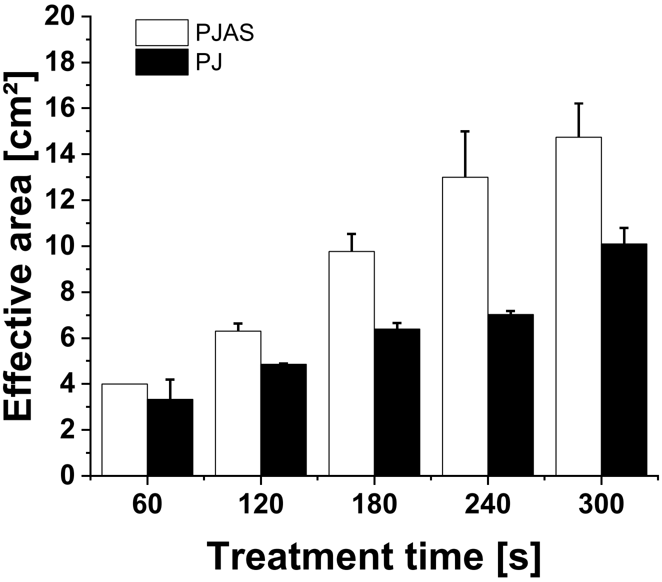

Figure 10. Average effective area (cm^2) (as arithmetic mean) with standard deviation (using Bessel's correction) for treatment times of 60 s, 120 s, 180 s, 240 s and 300 s for the PJAS II- 2 at 2.5 slm and PJ ($n = 3$).

increasing treatment time, the gaps between the spots were closed and the effective area resulted in an oval form. Whilst the affected areas rose with longer treatment time, the effect did not increase in a linear manner. Although, the four nozzles of the PJAS resulted in greater areas in comparison to the PJ with one nozzle, the increment was below a factor of 2 (figure 10). It must be considered that geometrical differences between these two devices, together with the varying flow rate and the variations resulting from this, have probably contributed to this non-linear change of the area.

After 60 s effective areas of 3 cm^2 to 4 cm^2 for both devices were detected, followed by an increase to almost 16 cm^2 (factor of 4) for the PJAS after 300 s. For the PJ, the area increased by almost a factor of 3 with a total area of up to 11 cm^2 after 300 s.

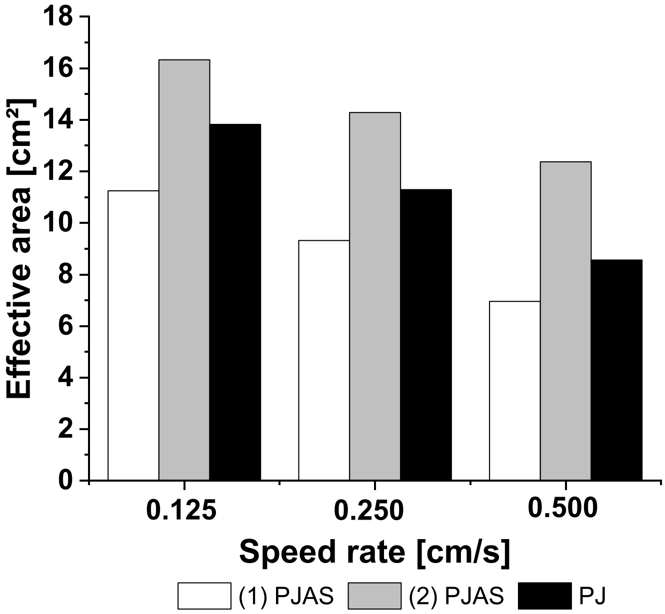

Figure 11. Effective area (cm^2) for the (1) PJAS II-2 (single movement, 2.5 slm), (2) PJAS II-2 (four movements, 2.5 slm) and PJ at three speed rates: 0.125 cm s^{-1}, 0.250 cm s^{-1} and 0.500 cm s^{-1} ($n = 1$).

The presented measurements were carried out from a fixed position of the devices. For this purpose, in the application of wounds or on the skin, the device will be moved across the surface to be treated. In order to mimic application conditions, the next experiments were carried out using a *XYZ* translation stage to perform a moving pattern, intended to simulate application conditions. Therefore, a pattern of four parallel lines was defined (see 2.3.2, figure 2). This movement pattern serves as the basis for both devices, the PJAS and the PJ.

The PJAS with four nozzles passed the plate in one movement, resulting in four times larger treatment area compared to one movement of the PJ. The PJ with only one nozzle had to be run four times to treat the same area as the PJAS.

Two scenarios were tested to evaluate the resulting effective treatment areas of the PJAS and the PJ.

(1) The PJAS was moved once (simulating a single four-line treatment).
(2) The PJAS was moved four times, equaling the same treatment time needed for four single movements of the PJ.

Additionally, to consider the impact of different speed rates, three different setups were tested for both devices and both scenarios: 0.125 cm s^{-1}, 0.250 cm s^{-1} and 0.500 cm s^{-1} (figure 11).

Regarding scenario (1), a single movement of the PJAS resulted in a smaller effective area of 11 cm^2 than for the four-line PJ treatment with 16 cm^2 for the speed rate of 0.125 cm^2. The evaluation of a single movement of the PJAS (see (1) PJAS, figure 11) shows a reduction from 11 cm^2 to 7 cm^2 at different velocities, whereas the effective area of the four movement PJAS treatment (scenario (2)) exceeded the four-line

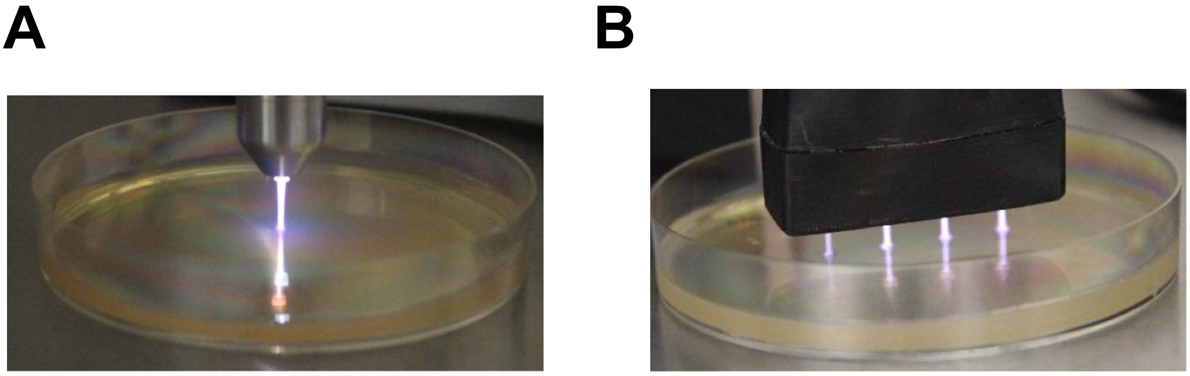

Figure 12. Effluent of the PJ (A) and of the PJAS II-2 (B).

PJ treatment due to the higher treatment time. Additionally, for PJAS (2) the area decreased from 16 cm^2 to 12 cm^2 (with an increasing speed rate) leading to a 1.3-fold reduction. The effective area of the PJ was slightly more reduced from 13.5 cm^2 to 8 cm^2 depending on the speed rate being 1.7 times lower.

For both devices the highest effective areas were stated at the lowest speed rate of 0.125 cm s^{-1} with a decrease at 0.250 cm s^{-1} and the smallest area for 0.500 cm s^{-1}.

A higher effective area results from the increase in the total treatment time for the lowest speed rate. Thus, the lowest speed rate (0.125 cm s^{-1}) represents a four times longer treatment duration in contrast to the highest speed rate (0.500 cm s^{-1}). However, the effective area did not decrease linearly, probably also for different geometries, which led to a deviating distribution of RONS.

For all tested setups it was observed that by a four-times faster treatment the effective area was maximally reduced by a factor of 1.7. In contrast, repeated treatment was beneficial. The effective area for PJAS (scenario 2) increased when the amount of treatments was raised. Exemplarily, an increase of 31% (from 11 cm^2 up to 16 cm^2) was achieved at the lowest speed (0.125 cm s^{-1}). However, the relatively low differences in the effective area for the various speed rates with a clear RONS production may also lead to the conclusion that the treatment velocity is not a crucial parameter. This is important to note because in the practical case of moving the device by handing over the area to be treated, an exact adherence to a certain speed is not realizable at all.

For practical applications it is therefore of importance to rather increase the number of treatments than to reduce the speed rate. Moreover, multiple applications reduce the risk of overseen areas being treated. We conclude that the resulting area of a single PJAS movement is not as effective as four movements of a PJ in the presently operated valve settings. Furthermore, the comparison of the same treatment times leads to larger areas of the PJAS device in contrast to the PJ.

The reason for the difference between both devices is their difference in the used architecture. The varying setup resulted in a change in argon gas flow rate. The flow rate of the PJAS was set to 2.5 slm, whereas the flow rate of the PJ was 5.0 slm. The shape also differs between the two devices due to the varying configurations. The effluent of the PJAS was rather narrow, while the PJ presented a broader effluent (figure 12).

In the case of the PJAS the production of reactive species is better distributed over a wider area due to the multiple nozzles, which allows a higher spread distribution of RONS with larger area coverage. These species cover a wider area while having a lower concentration due to the lower gas flow of 2.5 slm and the pulsed switching technology. Whereas for the PJ the species are emitted radially from one central spot, resulting in a purely radial region.

3.3. Cell viability

Cell viability experiments were carried out with human fibroblasts for treatment times of 10 s, 30 s, 60 s, 90 s and 120 s. Two controls (positive and negative) were applied. Negative control was performed with gas flow only, without plasma. The positive control whereas was conducted by a 0.00025% (7.5 µM) hydrogen peroxide (H_2O_2) solution as the acting agent (figure 13).

The results showed that for the PJAS the cell viability decreased during the treatment. Up to a maximum treatment time of 300 s, the cell viability steadily decreases to a value of approximately 30%. For negative control (gas flow without plasma) also a decrease over time was observed to be 62% . However, only a low decrease in cell vitality (to 70%) was observed for the positive control of a H_2O_2 concentration of 7.5 µM. This concentration was selected for the experiments due to the recommendations given in the DIN SPEC 91315:2014 [16]. Similarly, to our results, Mann *et al* [28] observed a decrease in cell viability after plasma treatment also for the PJ. While for the PJAS still viable cells were detected after maximum treatment time (figure 13), the authors described a complete elimination after PJ treatment. The comparison of a shorter treatment of 30 s resulted in a cell viability of about 80% for the PJAS and only 56% for the PJ. The different architecture and design of the devices might have affected the distribution of the gas flow and thereby the RONS generated by the plasma (see 3.2). While the PJ operates at a 50% duty cycle, the PJAS operates without burst but splits equally to each of the four channels, resulting in about one fourth of the plasma ignition per well. This may at least in part

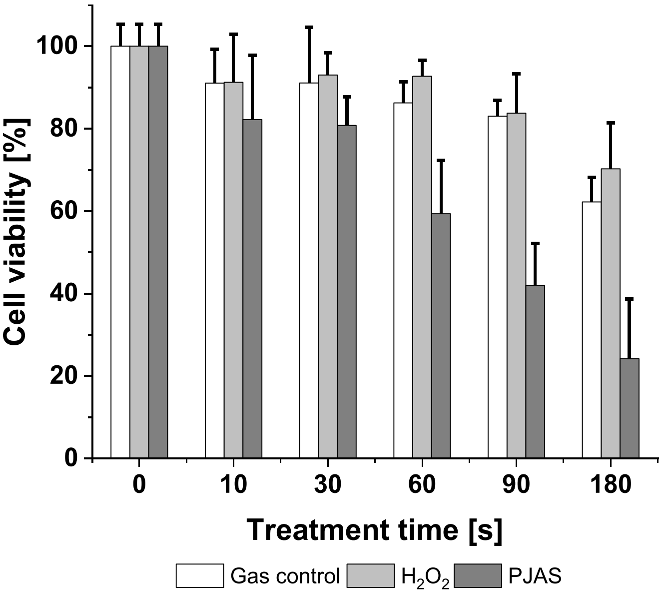

Figure 13. Average cell viability (%) (as arithmetic mean) with standard deviation (using Bessel's correction) of fibroblast cells (GM 00637) for 0 s (negative control/gas control) as well as 10 s, 30 s, 60 s, 120 s and 180 s treatment times, H_2O_2 (positive control) and the PJAS II-2 at 2.5 slm ($n = 3$).

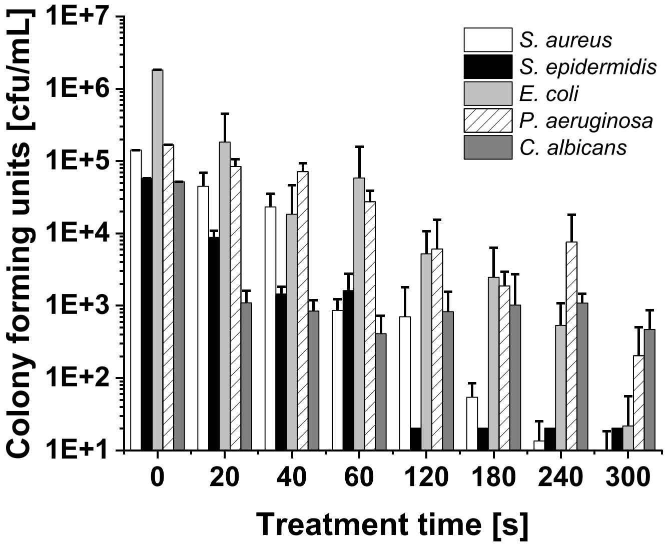

Figure 14. Average viable cell count described as colony-forming units (cfu ml^{-1}) (as arithmetic mean) with standard deviation (using Bessel's correction) in suspension assays for the PJAS II-2 at 2.5 slm for 0 s (gas control) as well as treatment times of 20 s, 40 s, 60 s, 120 s, 180 s, 240 s and 300 s for the microorganisms *staphylococcus aureus, staphylococcus epidermidis, Escherichia coli, pseudomonas aeruginosa* and *candida albicans* ($n = 3$).

be a reason for the higher cell viability after PJAS application compared to PJ.

Cell viability also decreased in gas control. The reason for this loss of vitality could be the selected volume for the experimental setup. The total volume per well was 300 µl. This low volume in the well is displaced by the gas inflow to such an extent that the cells come into contact with the air. The displacement may reduce the cell viability of the control and the plasma treatment. Thus, the actual cell viability after the plasma treatment under consideration of the control was 38% .

Summarizing the potential impacts, the interaction of reactive species, the gas flow and the relatively small volume could be decisive for this resulting decrease in the cell viability. In further studies more variations of controls should be made, which further determine the influence of reactive species and gas flow. Moreover, the volume that is treated should be varied, to exclude the impact of the gas flow rate on the liquid layer of the cells and, i.e., prevent a drying of the cells. Furthermore, indirect treatment of the buffer solution PBS added to the cells should also be considered to validate the influence on cell viability.

3.4. Antimicrobial efficacy in suspension

For the determination of the antimicrobial efficacy of the tested PJAS, petri dishes containing 5 ml of a suspension with the respective microorganisms *E. coli, S. aureus, P. aeruginosa, Staphylococcus epidermidis* and *C. albicans* were treated by the PJAS.

A decrease of cfu/mL with increasing treatment time was detected for the tested microorganisms. The correlation between treatment time and inactivation was dependent on the kind of microorganism, as described previously [27, 44, 45]. For all tested microorganisms, an antimicrobial effect of the PJAS was determined for treatment times from 20 s up to a maximum of 300 s. The overall inactivation varied from 2.1 orders of magnitude for *C. albicans* up to 4.9 for *E. coli* within a treatment time of 300 s (figure 14).

Remarkably, the inactivation was close to the detection limit for the three microorganisms *S. aureus, S. epidermidis* and *E. coli* within 300 s. Additionally, a reduction of over 2 orders of magnitude was detected for *P. aeruginosa* and *C. albicans*.

In sum, the $\log_{10}$-reduction values of the tested microorganisms and thereby the susceptibilities against PJAS from highest to lowest (after maximum treatment time) were ranked in the following order:

E. coli (4.9) > *S. aureus* (4.3) > *S. epidermidis* (3.5) > *P. aeruginosa* (2.9) > *C. albicans* (2.1)

The main reason for the varying results of the tested microorganisms is the susceptibility of the microorganisms toward plasma. Gram-positive (*S. aureus, S. epidermidis*) and gram-negative (*E. coli, P. aeruginosa*) microorganisms showed differences regarding their cell wall structure, which included variations regarding the cell wall thickness and composition [48], whereas the multi-layered cell wall of the yeast *C. albicans* has a far more complex structure [49]. However, the inactivation of *P. aeruginosa*, a gram-negative microorganism, was less than that for the other gram-negative one *E. coli*. This indicates that microorganisms can provide further defense mechanisms. It is striking that *P. aeruginosa* has green pigmentation, which could be reasonable for the

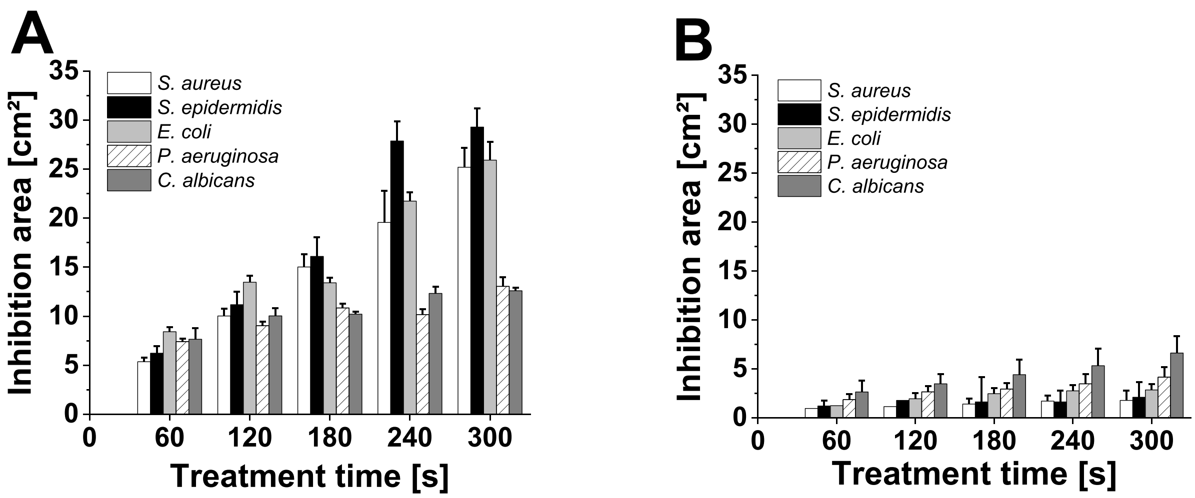

Figure 15. Average inhibition area (cm^2) (as arithmetic mean) with standard deviation (using Bessel's correction) for 0 s (gas control) as well as treatment times of 60 s, 120 s, 180 s, 240 s and 300 s for (A) PJAS II-2 at 2.5 slm and (B) PJ for the microorganisms *staphylococcus aureus, staphylococcus epidermidis, Escherichia coli, pseudomonas aeruginosa* and *candida albicans* ($n = 3$).

increased protection against plasma. The effect of pigmentation as a defending strategy was already evaluated in the past [50]. In addition, further defense strategies are conceivable [51, 52].

In particular, the reduced efficacy of *C. albicans* can be explained by their yeast characteristics. In differentiation of gram-positive and gram-negative microorganisms, yeasts possess a deviant composition and also avail other metabolisms. These structural and physiological differences make the yeast *C. albicans* more robust toward the influence of plasma [53, 54]. This can be seen for the PJAS, where the lowest inactivation was achieved for *C. albicans* ($\log_{10}$-reduction of 2.1). On the other hand studies of the PJ did not show any significant reduction ($\log_{10}$-reduction <0.5) for all five tested microorganisms [28]. Reasonable for these differing results could be the suspension volume treated by the plasma. Thus, this variation might explain the differences regarding the antimicrobial efficacy, due to the different distribution and amount of RONS in the suspension.

3.5. Antimicrobial efficacy on humid solid medium

Inhibition zone assays were implemented additionally to the suspension assays to evaluate the efficacy of the PJAS and to confirm the effective area predicted by the iodine-starch experiments.

As already described for the suspension assay, the inactivation was dependent on the respective treatment time, as well as the kind of microorganism and thereby their individual morphological and physiological characteristics (please refer to 3.4; figure 15). For treatment times from 60 s up to 300 s, an increase in the inhibition area was determined for the PJAS treatment ranging between 5 and 8 cm^2 after 60 s up to values of 12–30 cm^2 after 300 s (figure 15(A)). A far lower increase of the inhibition area with increasing treatment time was detected for the PJ reaching from approximately 1–2.5 cm^2 after 60 s to 2–7 cm^2 after 300 s (figure 15(B)).

In sum, the inhibition areas and thereby the susceptibilities against PJAS from highest to lowest (after maximum treatment time) were ranked in the following order: *S. epidermidis* (29.3 cm^2) > *E. coli* (25.9 cm^2) > *S. aureus* (25.2 cm^2) > *P. aeruginosa* (13.0 cm^2) > *C. albicans* (12.6 cm^2).

The highest inactivation was reached for the microorganisms *E. coli* and the two gram-positive microorganisms *S. aureus* and *S. epidermidis*, whereas *P. aeruginosa* and *C. albicans* showed a lower susceptibility. This trend was also shown by the suspension test (please refer to 3.4).

Contrastly, the inhibition areas after maximum treatment time (300 s) for the PJ (figure 15(B)) did not correlate to the inhibition areas of the PJAS (figure 15(A)). For the PJ it decreased in the following order: *C. albicans* (6.6 cm^2) > *P. aeruginosa* (4.2 cm^2) > *E. coli* (2.8 cm^2) > *S. epidermidis* (2.1 cm^2) > *S. aureus* (1.8 cm^2).

A comparison of the antimicrobial efficacy of PJAS and PJ showed a clear enhancement of the PJAS. Thus, the PJAS resulted in a higher inhibition area after 300 s for the treatment of *S. epidermidis* (30 cm^2 PJAS and 3 cm^2 PJ) by a factor of 10 and greater than 2 for *P. aeruginosa* (12 cm^2 PJAS and 6 cm^2 PJ after 300 s).

Next, the architectural deviations and therefore related changes in the distribution of RONS could be reasonable for the various inactivation results. The improved distribution of reactive species and further the antimicrobial efficacy was probably achieved by the multiple outlets and the gas flow dynamics supporting the inactivation of microorganisms. In practical conditions, such an optimized species distribution also reduces the burden for the user to homogeneously screen the whole wound geometry since an efficient treatment is applied over a larger area per design.

4. Conclusion

This study presents the development and evaluation of the PJAS, an innovative PJ array designed to enhance the treatment of larger areas (e.g., wounds) more efficiently than established PJ systems. By leveraging patented gas switching technology, the PJAS provides a scalable option to address safety requirements and increases in the treatment area and demonstrates up to tenfold improvements in antimicrobial efficacy.

Temperature profiling revealed safe regimes for the PJAS for all settings, maintaining skin-compatible levels below 40 °C. For PJAS I-1 and I-2, a distance beyond 9.5 mm for a 2.5 slm flow rate and beyond 7.5 mm for a 5 slm flow rate was found to reach temperatures that do not provide risks to tissues, while for PJAS II-2 these temperatures were detected for all investigated treatment distances. Flow dynamic impacts on the temperature over distance profile are discussed, implying an impact of the free jet structure on the PJ temperature. Furthermore, the gas switching technology induces stepwise temperature plateaus along the effluent of the PJAS. PLC measurements identified safe operation distances, ensuring compliance with medical safety standards. In particular, for the PLC, the PJAS II-2 can be found to define a safe treatment distance regime. Furthermore, ozone measurements showed values of the same order or below compared to the PJ.

Biological investigations showed a reduction in cell viability in human fibroblasts during *in vitro* tests. Thus, further studies, with variations on the used volume should be considered to reduce the impact of the control with just gas flow. Starch–potassium iodide experiments in motion and static indicated that the PJAS system oxidizes up to double the effective area compared to the PJ within identical treatment times, highlighting its enhanced capability to produce ROS. Suspension and inhibition zone tests showed that the PJAS system has an improved antimicrobial efficacy against the tested microorganisms in comparison with the PJ. As the susceptibility of plasma on the different microorganisms varied between both devices, further studies are needed.

The experimental data shows that the PJAS is a suitable device to treat large areas like wounds. However, further extensive studies are required to optimize operational parameters and nozzle numbers, ensure consistency, and comprehensively validate clinical safety and efficacy.

Data availability statement

The data that supports the findings of this study are openly available at the following URL/DOI: https://doi.org/10.34711/inptdat.876 [55].

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